

Exercises(2) (Continuous Optimization)

Exercise 1. Let A be an $m \times n$ real matrix and b a vector of $\mathbb{R}^m$. Consider the following problem

$$\inf_{v \in \mathbb{R}^n} \|Av - b\|$$

1. Prove that (P) admits at least a solution. Discuss the uniqueness according to the rank of A .
2. Show that u is a solution of (P) if and only if

$$A^t A u = 2A^t b.$$

Based on the rank of A , find back the results of the previous question.

3. For $\varepsilon > 0$, we define $J_\varepsilon(v) = \frac{1}{2}\|Av - b\|^2 + \varepsilon\|v\|^2$;
 - Show that there exists a unique $u_\varepsilon \in \mathbb{R}^n$ such that $J_\varepsilon(u_\varepsilon) = \inf\{J_\varepsilon(v), v \in \mathbb{R}^n\}$.
 - Precise the optimality condition satisfied by u_ε .
 - Show that $\|u_\varepsilon\| \leq \|u\|$ for all u solution of (P) . Deduce that (u_ε) converges, when ε goes to 0, to the solution of (P) of minimum norm.

Exercise 2. For $n \in \mathbb{N}^*$, consider the function: $J : \mathbb{R}^{n+2} \rightarrow \mathbb{R}$ defined by

$$J(v := (v_0, \dots, v_{n+1})) = h \sum_{i=0}^n \sqrt{1 + \left(\frac{v_{i+1} - v_i}{h}\right)^2},$$

where $h = \frac{1}{n+1}$

1. Give a geometrical interpretation of $J(v)$. Show that J is convex. Is it strictly convex?
2. We consider the problem of minimizing J on the set $K \subset \mathbb{R}^{n+2}$ defined by:

$$K := \{v = (v_0, \dots, v_{n+1}) \in \mathbb{R}^{n+2}, v_0 = 0, v_{n+1} = 1\}$$

Discuss the existence, the uniqueness, and characterize the (or) possible solution (s).